MARKING CRITERIA: Coursework – 80% (CW1)

	Level descriptors. *Note that these definitions are indicative of expected standards at each level, and may not be precise descriptors of the project submitted.*						
Category and marks weighting:	<30%	30-39%	40-49%	50-59%	60-69%	70-79%	80-100%
Project definition & planning (10%) "Scene setting" & defining the project aims & objectives, considers the problem domain and/or research question/task/problem (aims & objectives). Producing, adhering to and iteratively updating a structured project plan, considering time, resources, cost & ethics. Project vision is clear. Sprint plans map to backlog and aims. Releases planned every two weeks with appropriate plans and reviews.	Task totally unclear, undefined and/or irrelevant.	Inadequate task definition; very poorly defined.	Task definition under-developed; vague and/or illogical.	Task clarity and/or validity could improve.	Clear and sensibly defined task.	Task very well defined, explained and justified.	Perfect task definition; concise, compelling, SMART.
	Little/no evidence of having prepared or followed a structured project plan.	Inadequate project planning; tasks vague and/or illogically planned.	Some project planning and logic but needs much more thought, detail and/or clarity.	Project mostly well planned & managed. Some questionable logic and/or may lack detail.	Project generally well planned and managed.	Clear and logical project planning; dynamic response to setbacks/new discoveries.	Professional project management throughout all stages of the project.
Context review & subject knowledge (15%) Presenting a review of published literature that is relevant to this study with appropriate critical comment. OR presents a review of context of problem with appropriate critical comment. Using correct and appropriate citations throughout the report.	Little/no review of appropriate literature.	Minimal literature review. No critical comment; serious gaps/omissions.	Some literature review; little attempt at critical comment. Large gaps and omissions.	A range of relevant literature reviewed; some omissions. Some valid critical comment.	Appropriate breadth & depth to literature review; minor gaps. Good critical comment.	Comprehensive and very well written literature review with valid critical review throughout.	Comprehensive literature review: peer reviewed journal standard.

Showing a clear understanding of relevant subject matter throughout the project. Legal, Social, Ethical and Professional issues clearly and appropriately discussed along with appropriate critical comment and authoritative citations.	Missing/irrelevant list of references. Absence of valid citations throughout report.	Superficial reference list and/or predominately incorrect/inadequate citations.	Reference list present. Major presentation issues and/or poor citation style.	Reference list and/or citations could be more clearly/correctly presented.	Most citations and reference list entries correctly presented; some minor issues.	Highly competent use of citations and reference list.	Extensive list of references and citation style: peer reviewed journal standard.
	Little/No evidence of understanding of relevant subject matter throughout the project.	Inadequate understanding of relevant subject matter; confused &/or lacking depth.	Some understanding of subject matter with errors &/or poor conceptual framework.	Reasonable understanding of subject matter; minor errors. May require more depth.	Good understanding of subject matter with a clear conceptual framework.	Excellent understanding of subject matter beyond the level of taught modules.	Expert understanding of subject matter with rigorous attention to relevant detail.
	Little or no evidence of understanding of appropriate LSEP issues.	Inadequate understanding of relevant LSEP issues. Vague mentions made with little demonstrable understanding	Some understanding of LSEP issues with errors &/or poor conceptual framework.	Reasonable understanding of LSEP issues; minor errors. May require more depth.	Good understanding of LSEP issues with a clear conceptual framework.	Excellent understanding of LSEP issues beyond the level of taught modules.	Expert understanding of LSEP issues with rigorous attention to relevant detail.
Project methodology and implementation (50%) Defining an appropriate methodology and discussion of alternatives. Implementing the methodology to a depth appropriate for a 40 credit module.	Absence of anything that could reasonably be called a methodology.	Little/no justification for methodology &/or inappropriate methodology.	Some lack of logic/depth in justifying methodology. Major flaws in method selected.	Reasonable justification & selection of methodology with some flaws/errors/omissions.	Sensible, justified methodology selection with minor limitations.	Appropriate, well justified methodology selected. Good awareness of limitations.	Expertly justified; optimum methodology selected with due regard for limitations.

Demonstrating appropriate skills in the implementation of that methodology; skills depend on type of project (experimental, creative, mathematical, computational, etc.) Implementation of agile artifacts match proposed plans earlier or deviation from plan discussed appropriately. Implementation of code at appropriate level with demonstration of good software engineering principles. eg. DRY, YAGNI, SOLID.	Little or no relevant work done to achieve project aims & objectives.	Minimal relevant work done to achieve project aims & objectives.	Some relevant work but far short of that expected in the time allocated for 40 credits.	Project engagement &/or depth of coverage of task could improve.	Good project engagement and coverage of the task.	Good depth of coverage of the task, showing a high level of project engagement.	Comprehensive coverage of a highly demanding task. Very high level of engagement.
	Little or no evidence of relevant skills in project implementation.	Poor skills in implementing the methodology - incorrect &/or very confused.	Some skill in implementing the methodology, but with errors &/or confusion.	Skill in most areas of methodology implementation - some issues/errors.	Competent implementation of methodology with minor issues/errors.	Highly skilled implementation of methodology (far beyond the level of taught modules).	Expert level of skill in all relevant areas clearly evident throughout.
	Little or no evidence of implementation of agile project management	Minimal relevant work done to implement agile.	Some relevant work but far short of that expected for an agile project	Agile engagement &/or depth of coverage of task could improve.	Good agile application and coverage of the task.	Good depth of coverage of agile, high level of engagement with theory leading to good implementation. Agile worthy of commercial environment	Comprehensive coverage of a highly demanding agile implementation. Very high level of engagement.

	Little or no evidence of coding skills in project implementation.	Poor skills in implementing code - incorrect &/or very confused.	Some skill in implementing the software, but with errors &/or confusion. Application provides more functionality than log in and registration.	Skill in most areas of software implementation - some issues/errors. Implementation of moderate complexity with suitable functionality demonstrated.	Competent implementation of software with minor issues/errors. Application is of suitable complexity, has appropriate storage for any data, architecture is not monolithic but demonstrates interactions between levels and/or layers of software.	Highly skilled implementation of software (far beyond the level of taught modules). Application and architecture have good complexity, and good quality software engineering. Data storage if required is good, application goes way beyond a form of data storage with front end.	Expert level of skill in all relevant areas clearly evident throughout. Software is of commercial quality and could be implemented in real world situation with very little modification. OR research of quality that could easily lead to publication.
Critical evaluation & conclusions (15%) Appropriate mathematical/statistical methods to process & present data if appropriate. Software testing, verification and validation appropriate. Discussion/critical evaluation of results. Drawing the results together to form clear conclusions	A complete absence of appropriate data analysis.	Fundamentally flawed or inappropriate data analysis/processing.	Some relevant data analysis/processing with major issues/errors.	Appropriate data analysis/processing with some issues/errors.	Competent data analysis/processing with minor issues/errors.	Highly competent data processing/analysis & treatment of uncertainty.	Expert mathematical data processing; skillful error/sensitivity analysis.

linked to the project aims and objectives (quantitative if appropriate). Recommendations for further work. Taking feedback and findings from events and applying this to the project.	Project is devoid of appropriate testing plan	Poor skills in applying testing, incorrect &/or very confused	Some relevant testing applied. V&V superficial, sparse &/or often flawed	Appropriate testing in place but with some omissions, issues &/or errors	Competent testing plan in place. Appropriate Validation and Verification approach in place.	Highly competent testing regime in place both in plan and implementation. Shows a deep understanding of testing above and beyond taught modules.	Expert testing plans and implementations in place, could be appropriate for commercial application with very little modification.
	Project is devoid of critical analysis and evaluation.	Poor critical awareness showing little understanding of project results.	Critical evaluation is superficial, sparse and/or often flawed.	Appropriate critical evaluation in some areas with some omissions, issues and/or errors.	Competent critical evaluation in most areas of the project.	Highly competent critical awareness showing a good understanding of results.	Expert critical analysis throughout, showing deep understanding of results.
	Absent/irrelevant conclusions.	Inadequate/unjustified conclusions.	Conclusions vague and/or largely unjustified.	Relevant conclusions. Accuracy, evidence and/or clarity could improve.	Logical conclusions predominantly evidence-based and clearly presented.	Appropriate, well presented and well justified conclusions.	Clear, concise and fully quantitively justified conclusions.
Structure and presentation (10%) Structuring the report such that it is easy to follow, with information delivered in the right order, to an appropriate level of detail.	Little or no coherent report structure.	Structure lacks logic - rather "thrown together".	Some structure but disjointed/confusing.	Structure reasonable but could be easier to follow.	Sensible structure with minor issues/errors.	Excellent; clear and logical structure.	Faultless structure - perfectly presented.

Using appropriate grammar and language. Clearly presenting any mathematical work. Using appropriate, clear figures, images and graphs with correct labels, units, titles, etc.	Writing &/or mathematical notation incomprehensible.	Inappropriate written work and/or mathematical notation.	Poor literacy and/or mathematical notation.	Mainly appropriate style of writing and mathematical presentation - could improve.	Clear style of writing and mathematical presentation.	Lucid style of writing. Clear, unambiguous mathematical presentation.	Literacy/mathematical presentation: peer reviewed journal standard.
	Images/graphs/figs sparse, illegible and/or irrelevant.	Images/graphs/figs do not convey required information.	Most images/graphs/figs convey required information but may lack clarity and/or contain errors.	Mainly appropriate images/ graphs/figs - aesthetics and/or labelling could improve.	Most images/graphs/figs of high standard; occasional minor errors/issues.	Images/graphs/figs of high standard, clearly conveying all required information.	Creative images/graphs/figs; peer reviewed journal standard.